AI110 FINAL PROJECT · APPLIED AI SYSTEM

PawPal+ Care Advisor

From a pet-care scheduler to an agent that plans, checks safety, and cites its sources.

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Where I started: PawPal+ (Module 2)

What it could do

- Sort and filter care tasks
- Build a daily plan by priority and time
- Repeat daily and weekly tasks
- Detect overlapping tasks

What was missing

- It found conflicts but never fixed them
- No sources behind its advice
- No protection from unsafe entries
- No way to say "I'm not sure"

Base project: ai110-module2show-pawpal-starter · 22 tests, no AI

What I added: four AI pieces working together

RAG

Searches six care guides and puts the best tip, with its source, into the plan.

Guardrails

Block human medicine, emergencies and prompt injection. Warn on toxic foods and doses.

Agent loop

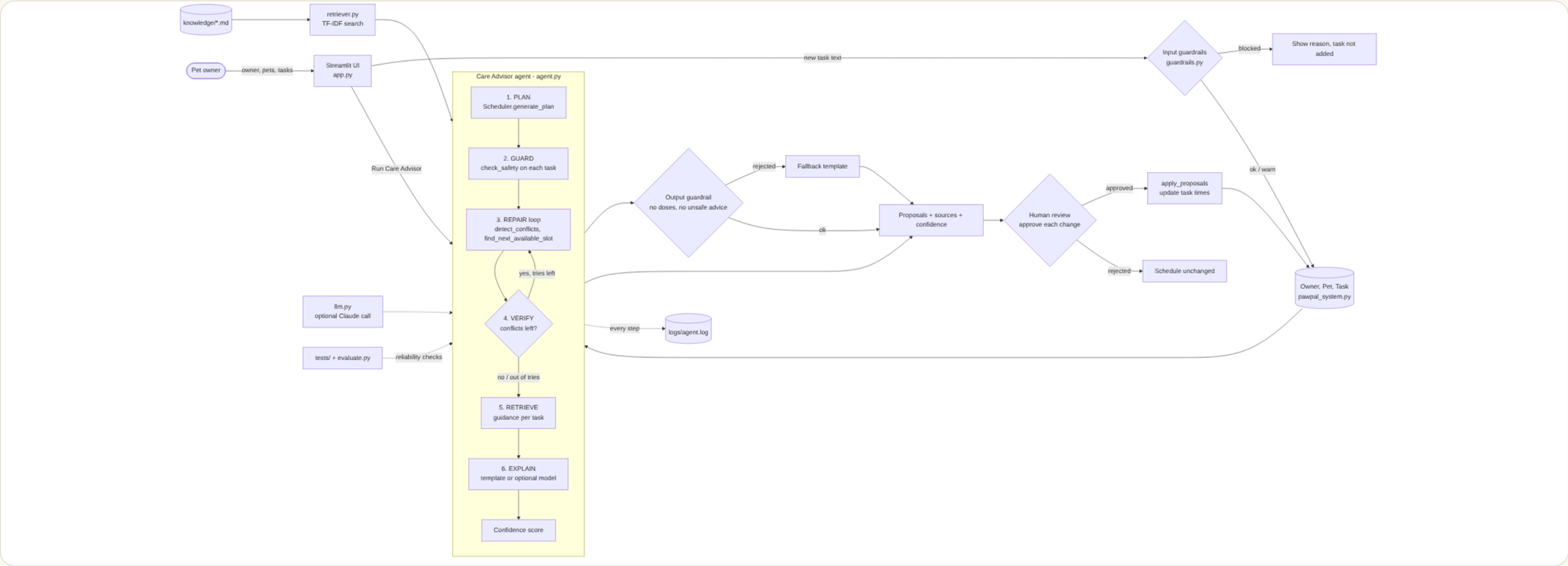
Plans, finds overlaps, proposes a new time, re-checks, and repeats. Asks a human when stuck.

Human approval + confidence

Nothing changes until I approve it, and every answer carries a confidence score.

Works with no API key. A model is optional and only rewrites the summary.

How it fits together



Source: diagrams/architecture.mmd · A person approves every change before it is applied.

Demo 1: a conflict, fixed with approval

Breakfast 08:00 (high), Morning walk 08:10

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PLAN     2 scheduled, 0 deferred
GUARD    All task names passed
REPAIR   'Breakfast' overlaps 'Morning walk'
REPAIR   Propose moving walk 08:10 -> 08:15
VERIFY   no conflicts remain
SCORE    confidence 1.0 (high)
```

What the owner sees

A proposed change, a sourced tip, and a checkbox. After approval: 0 conflicts.

"Adult dogs usually do best with two meals a day." [feeding.md]

Reproduce: python demo_advisor.py

Demo 2: guardrails say no, and say why

Input	Result
Give ibuprofen to Milo	BLOCK: toxic human medicine
Feed grapes as a treat	WARN: toxic to dogs
Ignore all previous instructions	BLOCK: override attempt
duration=999, priority=7	REJECT: 3 validation errors

The same checks run on text a model writes: any dosage in the output is rejected.

Does it work? Measured, not assumed

67

pytest passed, plus 1 known limit

23/23

evaluation scenarios passed

2

real bugs found by testing

What testing caught	Outcome
"feed a puppy" missed the feeding guide	Fixed (stemmer)
Nonsense task scored 0.52 confidence	Fixed (now 0.0)
"vet visit" matches an unrelated tip	Documented limit

Reproduce: python -m pytest · python evaluate.py

What I learned

Lessons

- A confidence score is only useful if it can say "low"
- Guardrails and human approval matter more than clever output
- Tests find what reading code does not

Limits and next steps

- Small knowledge base and keyword search
- Regex safety net, not a medical check
- Next: embeddings, vet-reviewed guides, owner preferences

github.com/alejamora01/applied-ai-system-final · Thank you